05. FAIRNESS METRICS & TESTING

~8 min

CONTENT

Demographic parity, Equalized odds, Predictive parity, Disparate impact ratio

Fairness metrics: Practical measurements for fairness. Demographic parity: Equal acceptance rate across groups. Equalized odds: Equal true positive rates across groups. Predictive parity: When model predicts positive, same confidence across groups. Disparate impact ratio: How much worse is one group treated? Measure: 80% rule (acceptance rate 80% as good for protected group). Testing process: Compute metrics on test set by demographic group. If metrics differ significantly, bias detected. Fix: Retrain with fairness constraint, adjust thresholds, or use fairness-aware algorithms. Real hiring example: 85% acceptance rate for men, 72% for women = disparate impact ratio 72/85 = 85% (fails 80% rule). Solution: Retrain with fairness constraint to achieve ~80% ratio.

06. RED TEAMING & ADVERSARIAL TESTING

~8 min

CONTENT

Deliberately breaking system, Edge cases, Adversarial examples, Stress testing

Red teaming: Deliberately try to break your system. Find failure modes before users do. Process: (1) Brainstorm: What could go wrong? Edge cases? Adversarial inputs? (2) Test: Try to break system. (3) Document: Record failures. (4) Fix: Address vulnerabilities. Edge cases for hiring: Very old candidates (70+ years, model hasn't seen this)? Career changers (no traditional path)? Military (classified companies not in training data)? Immigrants (credentials from other countries)? Non-traditional education (bootcamps, online). Test each: Does model handle fairly? Adversarial examples: Crafted inputs designed to fool model. Example: Resume optimized to game the system. Stress testing: What if input is extreme? 50 years experience (unrealistic but possible)? 100 job changes (weird but possible)? Stress test ensures robustness. Real team: Hire external red teamers (people outside your team) to break your system. They find things internal teams miss. Cost: $50-200K. Value: Prevents billion-dollar failures. Red team checklist: Demographics (test each demographic), Edge cases (unusual inputs), Adversarial (crafted malicious inputs), Scale (extreme values), Combination (multiple edge cases together).

07. INTERPRETABILITY & EXPLAINABILITY

~8 min

CONTENT

Black box to explainable, Feature importance, SHAP values, Transparency

Interpretability: Can you understand why AI made decision? User needs explanation. Legal requirement: EU AI Act requires explanations for high-stakes decisions. "Your resume didn't match" is not enough. Users want: "Why specifically didn't I match? Which skills missing? How can I improve?" Black box problem: Neural networks are inherently hard to interpret. Billions of weights, millions of interactions. Can't easily explain decision. Solutions: (1) Simpler models: Decision trees, linear models (interpretable by default). Tradeoff: Less accuracy. (2) Post-hoc explanations: Train complex model, then build explainability layer on top. Feature importance: Which features matter for decision? Example: Resume evaluation model. Feature importance says: Experience (40%), Education (25%), Skills (20%), Other (15%). User sees: "Experience is most important for your role." SHAP values: Shapley Additive exPlanations. Game theory approach to attribution. For each feature: How much did it contribute to final prediction? Example: SHAP values for candidate: Experience +0.3, Education -0.1, Skills +0.15, Certifications +0.05. Result: Candidate: "My experience helped most, education slightly hurt, skills and certs helped a bit." Transparency: Show model reasoning. Example: "For this role, we value: 1. Technical skills (60%), 2. Relevant experience (30%), 3. Communication (10%). Your profile scores: Technical 8/10, Experience 6/10, Communication 7/10. Total score: 7.1/10." User understands scoring. For hiring: Regulatory and ethical requirement. Candidates have right to explanation under many laws.

08. ROBUSTNESS & EDGE CASES

~8 min

CONTENT

Stress testing, Out-of-distribution data, Distribution shift, Graceful degradation

Robustness: Model works reliably even on unusual inputs. Edge cases: Rare but important scenarios. Example hiring: What if someone has 70 years experience? (unrealistic) What if they have 50 job changes? (unusual pattern) What if credentials are from obscure institution? (out of distribution) Distribution shift: Model trained on one distribution, deployed on different distribution. Example: Model trained on current job market. Market changes (recession, boom, new skills). Model fails because distribution shifted. Detection: Monitor model performance over time. If accuracy drops, likely distribution shift. Response: Retrain on new data. Graceful degradation: When uncertain, admit uncertainty. Example: Resume very different from training data. Model confidence 40% (low). System says: "This resume is unusual. Recommend human review." Better than: Model guesses and gets it wrong. Stress testing: Test on extreme values. Salary expectation: 1000x average? Commute time: 500 miles? Years of experience: 100? Negative skills (system says bad at something?). Test: Does model handle or fail gracefully? Robustness techniques: (1) Data augmentation: Add variations to training data so model sees diversity. (2) Adversarial training: Deliberately make data hard, train model to handle it. (3) Ensemble models: Multiple models vote, more robust. (4) Uncertainty quantification: Model says confidence level. (5) Graceful degradation: Fallback to human when uncertain.

09. REAL-WORLD FAILURES & LESSONS

~8 min

CONTENT

What went wrong, Why it went wrong, How to prevent, Regulatory response

FAILURE 1: Microsoft TAY CHATBOT. What: Microsoft deployed AI chatbot on Twitter. After hours, it became extremely racist/sexist. Why: Users deliberately fed it offensive content. AI learned from interactions. No content filter. No monitoring. Lesson: Monitor AI continuously. Have content filters. Don't deploy directly to adversarial environment. FAILURE 2: PREDICTIM RISK ASSESSMENT (Criminal Justice). What: AI system predicted criminal recidivism. Used by courts for sentencing. Found to show racial bias. Why: Training data: Historical criminal justice data which reflects decades of racial disparities. Model learned these patterns. Lesson: Be very careful with historical data that reflects past discrimination. FAILURE 3: GOOGLE PHOTOS LABELING BIAS. What: Google Photos AI labeled photos of Black people as "gorillas." Why: Training data had few images of Black people. Model overfit to gorilla images. Lesson: Diverse, representative training data essential. FAILURE 4: BIAS IN CRIMINAL RISK ALGORITHM. What: Algorithm called COMPAS used for parole decisions. Study showed it was biased against Black defendants. Why: Historical disparities in criminal data. Algorithm learned systemic bias. Lesson: Bias in criminal justice AI has life consequences. Deserves extra scrutiny. FAILURE 5: GENDER BIAS IN GOOGLE TRANSLATE. What: Google Translate defaulted to male pronouns for most professions, female for nurse. Why: Training data reflected gender stereotypes. Lesson: Even translation has bias. Pervasive in language data. PATTERNS: All failures had warning signs pre-deployment. All could have been caught with proper testing. All had severe consequences (reputation, legal, human). REGULATORY RESPONSE: EU AI Act: High-risk AI requires human oversight, auditing, documentation. GDPR: Right to explanation for automated decisions. Proposed regulations: Companies will need fairness documentation, bias testing logs. LESSONS FOR FUTURE: (1) Test for bias before deployment (mandatory). (2) Continuous monitoring post-deployment. (3) Diverse evaluation teams. (4) External audits. (5) Transparency with users.

10. SAFETY BEST PRACTICES

~5 min

CONTENT

Design for fairness from day one, Diverse teams, Continuous monitoring, Ethical review

Build safety into development process: (1) Data review: Is training data representative? Does it have known biases? (2) Fairness metrics: Decide which fairness definition matters. Set targets. (3) Diverse teams: Hire people from different backgrounds. Different perspectives catch bias. (4) Bias testing: Before deployment, test for bias. (5) External audit: Have independent people review. (6) Transparency: Document decisions, trade-offs, limitations. (7) Monitoring: After deployment, monitor for bias. (8) Feedback loops: Listen to users. If people complain about bias, investigate. Diverse teams critical: Homogeneous teams miss bias. Example: Mostly male team might not catch gender bias. Mostly one race might not catch racial bias. Different life experiences reveal different biases. Ethical review board: Before deployment, ethics board reviews system. Questions: Who is affected? Could this cause harm? Is it fair? Have we tested for bias? Do users understand limitations? Documentation: Document your process. What data did you use? What biases did you test for? What results? What limitations? This isn't academic. This is legal protection if something goes wrong. Continuous monitoring: Post-deployment: Monitor for bias emerging. If accuracy drops for one group, investigate. If fairness metrics change, investigate. User feedback: Most important. If users complain about bias, investigate thoroughly.

11. ALIGNMENT & VALUES

~5 min

CONTENT

Whose values does AI encode, Stakeholder input, Specification gaming, Long-term thinking

Alignment: Does AI system reflect your values? Problem: AI optimizes for metric. If metric is wrong, AI does wrong thing. Example: Recommendation algorithm optimizes for engagement. Engagement metric says "radical content is engaging." Algorithm recommends radical content. Result: Radicalization. Better metric: "User satisfaction + engagement." Algorithm balances both. Whose values: Who decides what's right? For hiring AI: Are you company? Candidates? Society? Different stakeholders have different values. Company: "Efficiency, cost." Candidate: "Fairness, opportunity." Society: "No discrimination." Specification gaming: AI finds loopholes in metric. Example: Optimize for "candidate acceptance rate among all applications." AI learns: "Reject bad candidates pre-screening, so acceptance rate high among those interviewed." Result: Less diversity in pipeline. Lesson: Metrics incentivize behavior. Think carefully about what you're incentivizing. Values alignment: What values should hiring AI encode? (1) Fairness (no discrimination)? (2) Merit (best person for job)? (3) Diversity (diverse backgrounds)? (4) Opportunity (give underrepresented groups chance)? These sometimes conflict. Merit vs diversity: Most qualified person often comes from privileged background (more opportunities). Diversifying might mean hiring less-qualified-on-paper but more-motivated person. Which value wins? Your call. But be explicit. Long-term thinking: Short term: Hire best on paper (high performance). Long term: Build diverse team (innovation, different perspectives, retention). Values affect long-term outcomes. Stakeholder input: Whose values matter? Probably all: (1) Company: Goals, constraints. (2) Candidates: Fairness, opportunity. (3) Society: No discrimination. (4) Employees: Team fit, values. Get input from all. Build system that reflects shared values.

12. LAB EXERCISE OVERVIEW

~3 min

CONTENT

Lab title: 'Design Safety & Fairness Testing' • Steps: Identify biases, Design tests, Plan monitoring, Create explainability

Your task: Design safety and fairness testing for your hiring AI. Identify potential biases. Design tests to catch them. Plan monitoring. Create explainability. Lab has 8 parts: (1) Identify potential biases (2 min), (2) Design fairness tests (2 min), (3) Plan red teaming (1 min), (4) Create explainability layer (2 min), (5) Define monitoring metrics (1 min), (6) Build ethical review process (1 min), (7) Document limitations (1 min), (8) Reflection (1 min). Deliverables: Bias identification document, Fairness test plan, Red team strategy, Explainability design, Monitoring dashboard design, Ethical review checklist, Limitation documentation, Reflection (2-3 paragraphs).

13. KEY TAKEAWAYS FROM SESSION 10

~5 min

CONTENT

6 Core Principles • Safety is business critical, Bias is everywhere, Fairness requires definition, Testing is mandatory, Transparency builds trust, Alignment takes effort

Takeaway One: Safety and fairness are business critical. Not ethical overhead. Amazon, healthcare, YouTube all learned this hard way. Takeaway Two: Bias is everywhere and subtle. Historical data, proxy variables, feedback loops. You must actively look for it. Takeaway Three: Fairness requires definition. Multiple valid definitions. Choose based on your values. Be explicit. Takeaway Four: Testing is mandatory. Before deployment: test for bias. After: monitor continuously. Takeaway Five: Transparency builds trust. Users deserve explanation. Legal requirement in many jurisdictions. Takeaway Six: Alignment takes effort. Whose values does system reflect? Probably multiple stakeholders. Find shared values.

14. COMING UP NEXT

~2 min

CONTENT

Dark blue background • Title: 'Session 11: Advanced Architectures' • Preview bullets

Session 11 explores cutting-edge architectures. Mixture of experts, multi-modal learning, attention mechanisms, Vision transformers. See you there.